

AI-Powered Multilingual Language Learning Chatbot

1. Project Overview

This project is an AI-powered language learning chatbot that supports real-time multilingual conversations. It allows users to select their native and target languages, choose their proficiency level, and interact with the bot in practical scenarios like ordering food or shopping. The system identifies language mistakes, provides corrections, and logs these for later review.

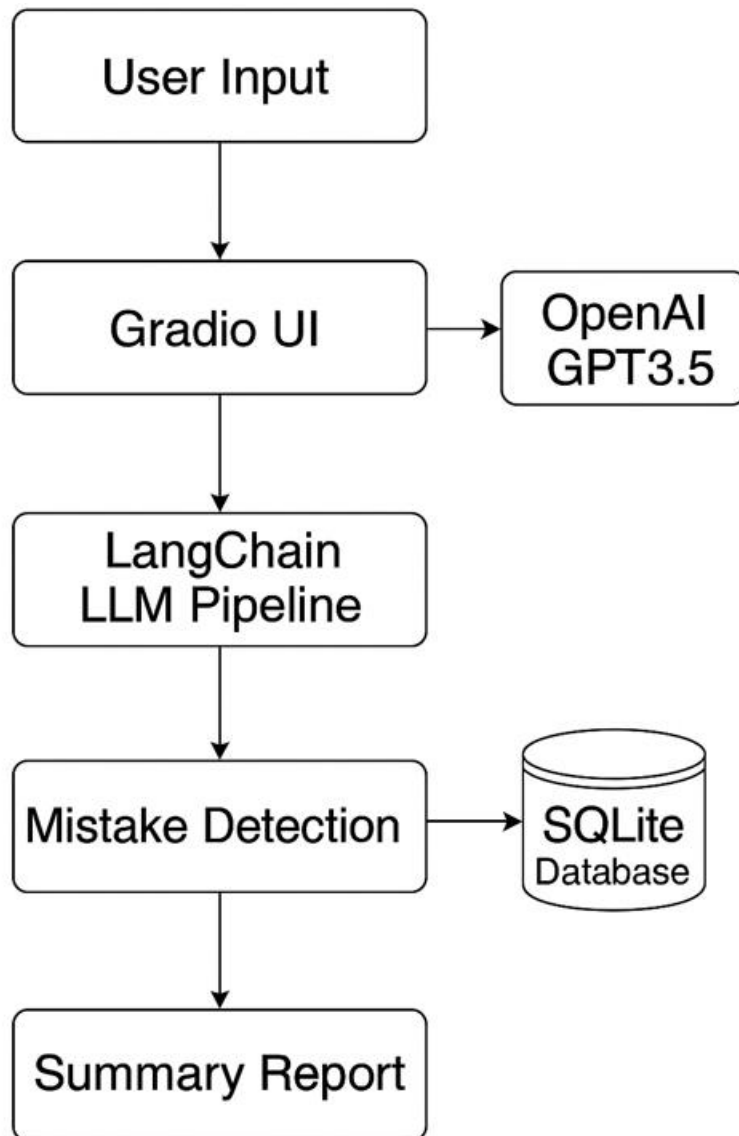
The chatbot is deployed on Hugging Face Spaces using a Gradio interface, integrated with OpenAI's GPT-3.5-turbo via LangChain, and logs user mistakes using SQLite.

2. System Architecture

The architecture follows a modular design with the following components:

- Gradio Interface: Provides a user-friendly web interface hosted on Hugging Face.
- LangChain + OpenAI GPT-3.5: Generates language-specific responses and handles correction logic.
- SQLite: Stores user input mistakes with timestamps for feedback and progress tracking.
- Google Drive (for Colab testing only): Used for persistent storage during development.

Architecture Diagram:



3. Explanation of Components

- User Input: User enters their native language, target language, proficiency level, and selects a scenario.
- Gradio UI: Captures input and displays conversation in a real-time chat format.
- LangChain LLM Pipeline: Uses OpenAI GPT-3.5 to simulate the conversation and detect mistakes.
- Mistake Detection: The model provides corrections if user input contains errors.
- SQLite Database: Logs all mistakes including timestamp, input, and correction.
- Summary Report: At the end, users can view a mistake summary stored in the database.

4. Technologies Used

- OpenAI GPT-3.5-turbo via LangChain
- Langchain-OpenAI, Langchain-Community
- Gradio
- Hugging Face Spaces
- SQLite3 (persistent storage)
- Google Colab (development and testing)
- Python 3.11

5. Deployment Details

The application is deployed using Hugging Face Spaces. The final app includes a real-time Gradio chatbot with multilingual support, mistake tracking, and post-session review capabilities. The URL Link for the deployed chatbot is below:

https://huggingface.co/spaces/TGChandu/Language_Learning_ChatBot

6. Conclusion

This multilingual language learning chatbot leverages the power of OpenAI's GPT-3.5 and LangChain to deliver an interactive and intelligent learning experience. The integration of Gradio for the user interface ensures accessibility and ease of use, while SQLite enables seamless tracking of user progress through mistake logging. By supporting a wide range of languages and real-life conversational scenarios, the system caters to learners at all proficiency levels. The architecture is modular, scalable, and secure. This project not only showcases the practical application of generative AI in education but also highlights an effective end-to-end deployment pipeline. It stands as a valuable tool for personalized and automated language acquisition.